

Theoretical Foundation

KEY THEORETICAL PROPERTIES OF ĆIRIĆ CONTRACTIONS

1. EXISTENCE OF FIXED POINT

- ✓ Every Ćirić contraction on a complete metric space has at least one fixed point
- ✓ Existence guaranteed without assuming continuity
- ✓ This is a major improvement over classical results

2. UNIQUENESS OF FIXED POINT

- ✓ The fixed point is unique
- ✓ Guaranteed by the contraction condition
- ✓ No ambiguity in the solution

3. CONVERGENCE OF ITERATIONS

- ✓ Picard iteration $x_{n+1} = Tx_n$ converges to the unique fixed point
- ✓ For any starting point $x_0 \in X$
- ✓ Convergence is often faster than Banach contractions

4. DISCONTINUITY

- ✓ Ćirić contractions need NOT be continuous
- ✓ This is a crucial generalization
- ✓ Example: Kannan's mapping (discontinuous but contractive)

5. CONTRACTION CONSTANT

- ✓ Ćirić contractions have a well-defined contraction ratio
- ✓ Determined by the parameters a_1, a_2, a_3, a_4, a_5
- ✓ Convergence rate depends on this ratio

6. COMPLETENESS REQUIREMENT

- ✓ Complete metric space condition is necessary
- ✓ Ensures Cauchy sequences converge
- ✓ Guarantees existence of fixed point